

DEMOS Consensus Mechanism (DAC) 2.0

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Decentralized Adaptive Consensus (DAC) 2.0

Overview

Decentralized Adaptive Consensus (DAC) 2.0 is a new consensus model that aims for exceptional efficiency, scalability, and performance in decentralized systems, while operating with zero transaction fees as a parachain. DAC 2.0 is based on principles of dynamic system adaptation, game theory, and collaborative sharding technology, and provides robust security.

Mechanism

1. Adaptive Collaborative Sharding with Global State Layer (ACS-GSL)

DAC 2.0 applies a sharding mechanism in which the blockchain is segmented into smaller partitions or "shards". Each shard processes its own transactions and smart contracts. The number of shards can adjust dynamically based on network load, promoting scalability and efficiency. Each shard works independently to enhance performance while maintaining consistency through a Global State Layer.

2. Proof-of-Longevity with Byzantine Fault Tolerance (PoL-BFT)

DAC 2.0 introduces a two layers consensus mechanism using the Proof-of-Longevity (PoL) model when updating the Global State Layer, where nodes are selected for the validation process based on the duration of their active and reliable participation in the network. This encourages long-term involvement and stability of the network. Inside shards, a Byzantine Fault Tolerant consensus is implemented to ensure robustness, security and scalability without overloading the whole chain

3. Game Theory Incentives (GTI)

Even in the absence of traditional rewards, DAC 2.0 incentivizes nodes via non-monetary means. Using game theory principles, nodes gain reputational scores based on their consistent participation, cooperation with other nodes, and the successful validation of transactions. Higher scores increase the likelihood of a node being chosen in the PoL selection process.

4. Dynamic Sharding

DAC 2.0 continually adapts to transaction volumes and dynamically adjusts the number and size of shards to maintain optimal performance at all times. Using a randomness layer powered by a Verifiable Delay Function, dynamic and consistent shard generation is ensured.

Advantages

1. Scalability: With Adaptive Collaborative Sharding, DAC 2.0 can process numerous transactions in parallel, significantly enhancing the network's capacity.
2. Security: The Distributed Ledger Verification method, combined with the Proof-of-Longevity model, ensures robust security against various attack vectors.
3. Efficiency: By dynamically adjusting the shard count based on network load, and enabling independent cross-shard validation through DLV, DAC 2.0 offers high efficiency.
4. Decentralization: The Game Theory Incentives maintain the decentralized ethos of the blockchain, encouraging nodes to cooperate for the benefit of the entire network.

Adaptive Collaborative Sharding with Global State Layer (ACS-GSL)

ACS-GSL is a mechanism that combines dynamic sharding with a global state layer for a scalable, efficient, and secure blockchain system.

Mechanism

1. **Adaptive Collaborative Sharding (ACS):** ACS divides the blockchain into smaller partitions or "shards". Each shard independently processes its transactions and executes smart contracts, improving the overall scalability and efficiency of the network. The number of shards dynamically adjusts according to the network load, ensuring optimized performance under varying conditions.
2. **Global State Layer (GSL):** The GSL is a universal ledger that maintains the state of the entire blockchain network. Each shard communicates its processed transactions to the GSL at the end of every epoch. The GSL then updates the global state based on these inputs, ensuring consistency across the network. Shards can access the GSL to fetch the updated state of any address across the blockchain network.
3. **Local Cache and Cache Invalidation:** To minimize frequent access to the GSL, each shard maintains a local cache of frequently accessed addresses or smart contracts. This cache is updated from the GSL at the end of every epoch or when a specific request requires fresh data. A smart cache invalidation strategy flags local data as stale based on certain triggers, prompting shards to update their local cache from the GSL.

Key Features and Solutions

1. **Decentralized GSL (D-GSL)**
To avoid a single point of failure and prevent the GSL from becoming a system bottleneck, the GSL is decentralized and mirrored across multiple nodes. This design distributes the load of updating and fetching data across the network, ensuring resilience and efficiency.
2. **Batch Processing and Differential Updates**
Shards batch their updates to the GSL, reducing the load and frequency of communications. Differential updates, wherein only changes from the previous state are communicated, further reduce the amount of data transferred and stored.

3. Robust Security

GSL's integrity is maintained using cryptographic techniques and consensus algorithms. Each change to the GSL is signed by the contributing shard and verified by other shards before being accepted, ensuring trust and reliability.

Advantages

1. Scalability: ACS-GSL enhances the network's capacity through parallel processing of transactions in each shard and efficient management of the global state.
2. Efficiency: The strategy reduces cross-shard communication and optimizes GSL updates, leading to a high level of efficiency.
3. Consistency: GSL maintains a consistent state of the blockchain across all shards, ensuring that all nodes agree on the state of the network.
4. Interoperability: GSL improves interoperability between shards, allowing any shard to access the state of any address across the blockchain network.

Proof-of-Longevity with Byzantine Fault Tolerance (PoL-BFT)

Proof-of-Longevity with Byzantine Fault Tolerance (PoL-BFT) is a two-tiered consensus mechanism that promotes network stability, efficiency, and security in a blockchain system. It integrates a Byzantine Fault Tolerance (BFT) consensus within each shard and uses Proof-of-Longevity (PoL) for proposing blocks to the Global State Layer (GSL).

The Two Consensus Tiers

1. Byzantine Fault Tolerance (BFT) within Shards

Within each shard, a BFT consensus mechanism is utilized. BFT allows for quick transaction validation and agreement within the shard, even in the presence of malicious or faulty nodes. It ensures efficient operation and robust security within each shard by requiring a supermajority (typically $2/3+1$) of nodes to agree on a new block before it is added to the shard's blockchain.

2. Proof-of-Longevity (PoL) for GSL

When proposing blocks to the GSL, the PoL mechanism is used. Validators for this process are selected based on their longevity score, which is determined by the length of their active and reliable participation within their respective shards. The likelihood of a node being chosen to propose the next block to the GSL is proportional to its longevity score. This design encourages nodes to maintain consistent, reliable participation in the network, enhancing overall stability and security.

Advantages

1. **Efficiency:** Using a BFT consensus mechanism within shards enables high-speed transaction processing and efficient intra-shard operation.
2. **Security:** The PoL consensus mechanism enhances security when proposing blocks to the GSL. By giving preference to nodes with long-standing, reliable participation, PoL minimizes the risk of malicious behavior and promotes the overall integrity of the blockchain.
3. **Scalability:** The PoL-BFT approach provides a scalable solution, capable of managing a high volume of transactions while maintaining the consistency and reliability of the GSL.
4. **Stability:** By rewarding long-term, consistent participation, the PoL mechanism encourages stability in the blockchain network.

Game Theory Incentives (GTI)

Game Theory Incentives (GTI) is a unique incentivization strategy that promotes network stability, cooperation, and effective performance in a zero gas fee blockchain system. Rather than relying on direct financial rewards, GTI utilizes principles from game theory to encourage nodes to act in the best interest of the network.

Design and Mechanism

1. Reputation Score

Each node is associated with a reputation score, which is a measure of the node's performance, reliability, and contribution to the network. The reputation score is dynamic and changes based on the node's activities. Successfully validating transactions, maintaining consistent uptime, and contributing to the security of the network increase the reputation score, while fraudulent activities or failures decrease it.

2. Probabilistic Selection

Nodes with higher reputation scores have a higher probability of being selected as validators in the BFT consensus within shards and as proposers in the PoL consensus for the GSL. This selection process is not deterministic; a node with a lower reputation score still has a chance of being selected, albeit smaller. This probabilistic approach maintains a balance between rewarding high-performing nodes and allowing newer or less-performing nodes to participate and improve.

3. Cooperative Behavior

The system encourages cooperative behavior among nodes. When nodes cooperate (for example, by correctly validating transactions, synchronizing with the network, and actively participating), they mutually benefit due to the increase in overall network performance, reliability, and security, which in turn leads to an increase in their reputation scores.

Advantages

1. **Encourages Active Participation:** GTI encourages nodes to maintain active and reliable participation in the network, as this increases their chances of being selected as validators or proposers.
2. **Enhances Security:** By rewarding correct behavior and penalizing fraudulent activities, GTI helps maintain the integrity and security of the network.
3. **Promotes Fairness:** The probabilistic selection process ensures all nodes, regardless of their reputation score, have a chance of participating in the consensus process.
4. **Improves Network Performance:** GTI encourages cooperation among nodes, leading to improved overall network performance.

Dynamic Sharding

Dynamic Sharding is a technique where the number of shards (subdivisions of the blockchain network) can dynamically adjust based on the network's transaction load and node distribution. This adaptability allows the blockchain network to maintain optimal performance under a variety of conditions, from low to high transaction demand.

Design and Mechanism

1. **Shard Creation and Dissolution:** When the transaction load or the number of nodes in the network increases significantly, the system creates new shards to distribute the load and maintain efficient processing times. Conversely, if the transaction load or node count decreases, some shards can be dissolved and their responsibilities redistributed to maintain resource efficiency.
2. **Load Balancing:** Dynamic Sharding also provides load balancing across the network. The system periodically evaluates the transaction load on each shard and reassigns nodes or reallocates transactions to balance the workload and optimize the processing speed.
3. **Node Allocation:** When new nodes join the network or existing nodes leave, the system reevaluates the distribution of nodes across the shards and reassigns nodes as needed to maintain a roughly equal number of nodes in each shard.

Advantages

1. **Scalability:** Dynamic Sharding enables the network to manage increased transaction loads by creating additional shards, significantly enhancing the network's scalability.
2. **Efficiency:** By balancing the workload and node distribution across shards, Dynamic Sharding ensures efficient transaction processing and optimal resource utilization.
3. **Resilience:** The ability to adjust the number of shards based on network conditions makes the blockchain network more resilient to fluctuations in transaction demand or node participation.
4. **Lower Latency:** By distributing transactions across multiple shards, Dynamic Sharding can help decrease transaction validation and block creation times, resulting in lower latency.

Shard Generation with a Verifiable Delay Function (VDF)

The process of shard generation using a Verifiable Delay Function (VDF) involves creating a shared, deterministic source of randomness to ensure all nodes generate the same shard configuration given the same network state.

Step 1 - Initiate Epoch

At the start of each epoch, the blockchain system triggers the VDF process. This process starts with the selection of an initial input, often referred to as the "seed". This seed could be derived from the hash of the last block of the previous epoch or any other deterministic value shared across the network.

Step 2 - Apply VDF

The selected seed is then input into the VDF. The VDF requires a specific amount of time to compute, producing a unique output after a predetermined delay. The output cannot be predicted before the computation is complete, ensuring security and unpredictability.

Step 3 - Verification and Distribution

Once the VDF has produced the output, this output can be quickly verified by all other nodes in the network. Upon verification, this output is propagated across the network. The fast verification property of the VDF ensures that the output is disseminated to all nodes nearly simultaneously.

Step 4 - Shard Generation

Each node in the network uses this output as the seed for shard generation. Since the seed is identical for all nodes, the shard generation process results in the same set of shards across the entire network. The nodes are then assigned to these shards based on the deterministic process using the same seed, ensuring uniform distribution and balanced workload.

Step 5 - Begin Epoch Operations

With the shards generated and nodes assigned, the blockchain network proceeds with its operations within the new epoch, processing transactions within each shard, and committing states to the Global State Layer.

The utilization of a VDF within the randomness layer provides a secure, verifiable source of randomness, enabling consistent shard generation across the network, enhancing security, and ensuring scalability.

Wrap up and recap

The newly proposed blockchain consensus mechanism presents a **robust**, **scalable**, and **innovative** solution, integrating features of Adaptive Collaborative Sharding with Global State Layer (ACS-GSL), Proof-of-Longevity with Byzantine Fault Tolerance (PoL-BFT), Game Theory Incentives (GTI), Dynamic Sharding, and a secure Randomness Layer powered by a Verifiable Delay Function (VDF).

Through ACS-GSL, the system efficiently manages transaction processing and data storage across various shards, enhancing scalability while ensuring a consistent view of the global state.

The PoL-BFT consensus model integrates a Byzantine Fault Tolerance consensus within each shard with a Proof-of-Longevity mechanism for global block proposals, optimizing both intra-shard and inter-shard operations.

The GTI framework encourages active, reliable participation and cooperative behavior among nodes, fostering network stability and security.

Dynamic Sharding enhances the system's adaptability, ensuring optimal performance and resource utilization regardless of network load. This, combined with the integration of a VDF-powered Randomness Layer, ensures secure, verifiable, and consistent shard generation, further contributing to the network's robustness and security. This innovative approach to generating shared randomness enables the system to maintain uniformity and fairness in the node distribution and workload allocation, further enhancing the network's scalability.

This novel blockchain system design showcases the potential of integrating various advanced concepts and technologies in building a blockchain network that is not only secure, robust, and efficient but also highly scalable and adaptable to varying network conditions.